



**Verified Carbon
Standard**

UMUT-II HYDROPOWER PLANT, TURKEY



Project title	Umut-II Hydropower Plant, Turkey
Project ID	1011
Crediting period	21-02-2024 to 20-02-2034
Original date of issue	12/06/2025
Most recent date of issue	12/06/2025
Version	04
VCS Standard Version	4.7
Prepared by	GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık Ve Ticaret A.Ş. M. Kemal Demirkol

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Audit History.....	6
1.3	Sectoral Scope and Project Type	7
1.4	Project Eligibility	7
1.5	Project Design	8
1.6	Project Proponent	8
1.7	Other Entities Involved in the Project	8
1.8	Ownership.....	8
1.9	Project Start Date	9
1.10	Project Crediting Period	9
1.11	Project Scale and Estimated GHG Emission Reductions or Removals	9
1.12	Description of the Project Activity	10
1.13	Project Location	12
1.14	Conditions Prior to Project Initiation	13
1.15	Compliance with Laws, Statutes and Other Regulatory Frameworks	14
1.16	Double Counting and Participation under Other GHG Programs	16
1.17	Double Claiming, Other Forms of Credit, and Scope 3 Emissions.....	16
1.18	Sustainable Development Contributions	17
1.19	Additional Information Relevant to the Project	17
2	SAFEGUARDS AND STAKEHOLDER ENGAGEMENT	18
2.1	Stakeholder Engagement and Consultation.....	18
2.2	Risks to Stakeholders and the Environment.....	22
2.3	Respect for Human Rights and Equity	25
2.4	Ecosystem Health	28
3	APPLICATION OF METHODOLOGY.....	30
3.1	Title and Reference of Methodology	30
3.2	Applicability of Methodology	31
3.3	Project Boundary	37
3.4	Baseline Scenario	42

3.5	Additionality	44
3.6	Methodology Deviations	45
4	QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS.....	45
4.1	Baseline Emissions	45
4.2	Project Emissions	46
4.3	Leakage Emissions	48
4.4	Estimated GHG Emission Reductions and Carbon Dioxide Removals	48
5	MONITORING	50
5.1	Data and Parameters Available at Validation	50
5.2	Data and Parameters Monitored.....	52
5.3	Monitoring Plan.....	55
	APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION.....	59
	APPENDIX X: <TITLE OF APPENDIX>	60

1 PROJECT DETAILS

1.1 Summary Description of the Project

“Umut-II Hydropower Plant, Turkey” (hereafter referred to as the “Project”) owned by RHG Enertürk Enerji Üretim ve Ticaret A.Ş. (hereafter referred to as the “Project Proponent”) is a large scale, run-of-river hydropower project located on the Karakuş River, Ordu, Türkiye.

RHG Enertürk Enerji Üretim ve Ticaret A.Ş. operates Umut-II Hydropower Plant, Türkiye with 24.450 MWe (25.305 MWm) run-of-river hydropower electricity plant (HEPP) is located on Karakuş River, Ordu, Türkiye. There are three turbines, each with 8.15 MWe/8.435 MWm capacity in Umut II HEPP. The expected lifetime of the project in accordance with the Generation License, is roughly 48 years, 8 months, and 4 days which corresponds to the duration of the license obtained from EMRA. Project is a grid connected renewable power generation project which adds electricity capacity from hydro power sources which otherwise would have been generated by the fossil fuel fired power plant. The main objective of the Project Proponent is to increase the power generating capacity in Türkiye by developing, constructing and operating renewable energy power plants with the support of carbon credits.

The purpose of the Project is to generate electrical power in an efficient, clean, reliable and sustainable way with utmost respect on social and environmental aspects and to reduce emissions by partially substituting the electricity supply of fossil fuel fired power plants in Türkiye. The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

The meters only measure gross electricity generation of Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP combined. Only data available from the project field is the combined gross electricity production of Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP. This is because Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP are both connected to the same grid connection point and only one meter is responsible to meter both plant's data.

Summary of information provided by the project proponent:

- Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP are connected to the same grid connection point and the metering is done with one meter.
- Gross electricity generation data provided by the project proponent shows three separate gross electricity production data set one belonging to Umut-I HEPP, one belonging to Umut-II HEPP, and one belonging to Umut-III HEPP. This does not indicate that the three different plants were

metered separately. But instead, the total value was rationed between Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP considering their plant capacities.

- EPIAŞ data stands for values for Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP combined.
- Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP operates under RHG Enertürk Enerji Üretim ve Ticaret A.Ş.

The project proponent states that a proportional constant is used to calculate gross electricity production of Umut-II HEPP alone and this constant can be determined by using the ratio between plant capacity of Umut-II HEPP and the total plant capacity of Umut HEPPs. The project capacity of Umut-I HEPP, Umut-II HEPP, and Umut-III are 5.800 MW, 24.450 MW and 12.000 MWe, respectively. To calculate the gross electricity production only for Umut-II HEPP, the combined gross electricity production of the three plants are multiplied by the capacity proportion of Umut II HEPP's within the total project capacity i.e., $24.450 / (5.800 + 24.450 + 12.000)$, which makes 0.5787.

As per Generation License, the average annual electricity generation for Umut HEPPs is 124,894 MWh/year. In this regard, for Umut II HEPP, the estimated annual electricity generation is 72,276 MWh/year ($0.5787 \times 124,894 \text{ MWh/year} = 72,276 \text{ MWh/year}$).

The project start date is 21/02/2014, as per Provisional Acceptance of Umut II HEPP. During this crediting period (2nd CP: 21/02/2024 – 20/02/2034), the annual generation and the corresponding emission reduction is estimated as 72,276 MWh/year and 33,013 tCO₂/year, respectively. Total electricity generation and emission reduction for this crediting period are 722,759 MWh and 330,130 tCO₂, respectively.

The project milestones are provided in the table below:

Table 1. Project implementation milestones

Milestone	Date
Generation License	01/05/2008
Environmental Impact Assessment Exemption ¹	13/05/2008
Environmental Impact Assessment Approval ²	17/12/2009
Commissioning of the Project (Project Start Date) ³	21/02/2014

¹ Please see the file named Original EIA Exemption Doc

² Due to change in capacity, an assessment is conducted according to EIA Report page 369

³ Please see the Provisional Acceptance of Umut II HEPP

Verdict for the continuation of Environmental Impact Assessment Exemption ⁴	06/07/2017
First Crediting Period (1 st CP)	21/02/2014 – 20/02/2024
First Monitoring Period (1 st CP 1 st MP)	21/02/2014 – 20/02/2020
Second Monitoring Period (1 st CP 2 nd MP)	21/02/2020 – 20/02/2024
Second Crediting Period	21/02/2024 – 20/02/2034

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
<i>Validation/Registration</i>	21-February-2014 to 20-February-2024 (1 st Crediting Period)	VCS	TÜV Rheinland Ltd.	Ten years
<i>Verification</i>	21-February-2014 to 20-February-2020 (1 st Crediting Period 1 st Monitoring Period)	VCS	Earthood Services Limited	Six years
<i>Verification</i>	21-February-2020 to 20-February-2024 (1 st Crediting Period 2 nd Monitoring Period)	VCS	Re-carbon	Four years
<i>Renewal of Crediting Period</i>	21-February-2024 to 20-February-2034 (2 nd Crediting Period)	VCS	Earthood Services Limited	Ten years

⁴ Please see the file named Verdict_06.07.2017

1.3 Sectoral Scope and Project Type

Sectoral scope⁵	Scope number 1: Energy industries (renewable / non-renewable sources)
Project activity type	Grid-connected electricity generation from renewable sources

1.4 Project Eligibility

1.4.1 General eligibility

According to the eligibility criteria set in VCS standard version 4.7,

1) Umut-II Hydropower Plant, Turkey, is a grid connected renewable power generation project that applies a methodology eligible under VCS Program. The project capacity is 24.450 MWe, which is higher than 15 MWe, which is the capacity limit of the small-scale projects. Project activity includes renewable power generation projects from hydro power sources, and the activity is supported by a methodology approved under a GHG Program i.e., CDM which is an active methodology in VCS Program. The project has its own hydromechanical equipment and building. Thus, as per paragraph 2.1.1 of the VCS Standard version 4.7, the project activity is included under the scope of the VCS Program.

2) The project start date is the date (21/02/2014) on which the project began generating GHG emission reductions or carbon dioxide removals as per VCS Standard version 4.7. As stated in paragraph 3.8.1 of the VCS Standard version 4.7, for non-AFOLU projects the validation shall be completed within two years of the project start date. The validation for Umut II HEPP is completed accordingly.

3) The applied methodology which is ACM0002 Version 22.0 is a CDM methodology, and it is an active methodology in VCS Program.

The project is situated in a country that is not classified as a Least Developed Country (LDC) by the United Nations and is therefore not included in Table 1 of the VCS Standard version 4.7. Nevertheless, the project has been successfully registered with the Verra registry, and its activity pertains to the Renewal of Crediting Period (RCP).

1.4.2 AFOLU project eligibility

This project is not an AFOLU project.

1.4.3 Transfer project eligibility

This is not a transfer project or Component Project Activity seeking registration. Thus, this section is NA.

⁵ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.5 Project Design

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

This is not a grouped project.

1.6 Project Proponent

Organization name	RHG Enertürk Enerji Üretim ve Ticaret A.Ş.
Contact person	Kutalmış Karacan
Title	-
Address	Huzur Mah. Cendere Cad. Skyland İstanbul B Blok Kat:16 No:114B/300 Sarıyer/İSTANBUL/TÜRKİYE
Telephone	+90 212 267 42 06
Email	kutalmis.karacan@enerturk.com

1.7 Other Entities Involved in the Project

Organization name	GTE Karbon Sürdürülebilir Enerji Eğt. Dan. ve Tic. A.Ş.
Role in the project	Project consultant
Contact person	M. Kemal Demirkol
Title	Director
Address	Mustafa Kemal Mah. 2111. Sok. No: 5 06530 Cankaya - Ankara – TÜRKİYE
Telephone	+90 312 514 63 63
Email	kemal.demirkol@gte.com.tr

It is to be noted that the project consultant Mavi Sürdürülebilir Kalkınma Proje ve Danışmanlık Hizmetleri Ltd. Sti. mentioned in the registered PD as other entities involved in the project has changed to GTE

Karbon Sürdürülebilir Enerji Eğitim Danışmanlık ve Ticaret A.Ş., as GTE was contracted for the first verification period services.

1.8 Ownership

The Project is owned by the RHG Enertürk Enerji Üretim ve Ticaret A.Ş. The generation license issued by the EMRA (Energy Market Regulatory Authority) names the Project proponent as the holder of title for this Project activity.

1.9 Project Start Date

Project start date	21/02/2014
Justification	Project started in 21/02/2014 ⁶ , which is the provisional acceptance for this project activity. According to VCS v4.7, the project start date is defined as the date on which “the project began reducing or removing GHG emissions”.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☐ Ten years, fixed ☒ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements) Ten years, twice renewable as per registered PD which refers to VCS Standard v3 paragraph 3.9.1.
Start and end date of first or fixed crediting period	21 February 2024 – 20 February 2034

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☒ < 300,000 tCO₂e/year (project)
☐ ≥ 300,000 tCO₂e/year (large project)

⁶ Provisional Acceptance of Umut II HEPP

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
21 February 2024 to 31 December 2024	28,413
01 January 2025 to 31 December 2025	33,013
01 January 2026 to 31 December 2026	33,013
01 January 2027 to 31 December 2027	33,013
01 January 2028 to 31 December 2028	33,013
01 January 2029 to 31 December 2029	33,013
01 January 2030 to 31 December 2030	33,013
01 January 2031 to 31 December 2031	33,013
01 January 2032 to 31 December 2032	33,013
01 January 2033 to 31 December 2033	33,013
01 January 2034 to 20 February 2034	4,600
Total estimated ERRs	330,130
Total number of years	10
Average annual ERRs	33,013

1.12 Description of the Project Activity

The Project involves the generation of electricity and sales of carbon credits with hydroelectric power. It will help the host country as a developing country to stimulate and commercialise the use of grid connected renewable energy technologies and markets.

It will make use of water from the Karakuş River tributaries. The Project employs 3 Francis-type turbines with 8.15 MWe capacity each, totalling to 24.45 MWe. The Project will be operated all year round, at reduced capacity in the summer because of diminishing water flow.

Table 2. Installed capacities of Umut I-II-III

Installed Capacity of Umut-I	2x3.037 MWm / 2x2.900 MWe = 6.074 MWm / 5.800 MWe
Installed Capacity of Umut-II	3x8.435 MWm / 3x8.150 MWe = 25.305 MWm / 24.450 MWe
Installed Capacity of Umut-III	3x4.255 MWm / 3x4.000 MWe = 12.765 MWm / 12.000 MWe

The technical properties of the Umut II HEPP are given below.

Table 3. Technical Properties of Umut-II Hydropower Plant, Türkiye

Location	Karakuş River, Ordu Province, Türkiye
Total Installed Capacity	24.45 MWe
Number Of Units	3
Turbine Capacity	3 x (8.435 MWm / 8.150 MWe)
Turbine Type	3 x Francis turbines
Average Annual Power Generation	72,276 MWh/year

The pressure of the water flow will drive the turbine blades, which will drive the coupled generators to produce electricity. The generated power will be used for internal consumption and the rest will be supplied to the national power grid. The generation license has been obtained on 01/05/2008 for 48 years, and the Project proponent also foresees a lifetime of roughly 48 years, 8 months and 4 days.

The Project will partially divert the water flow of Karakus River and its tributaries to multiple diversion weirs to regulate the water flow and convey it to the powerhouse. The Project has the sole purpose of energy generation and is a run-of-river type hydropower plant.

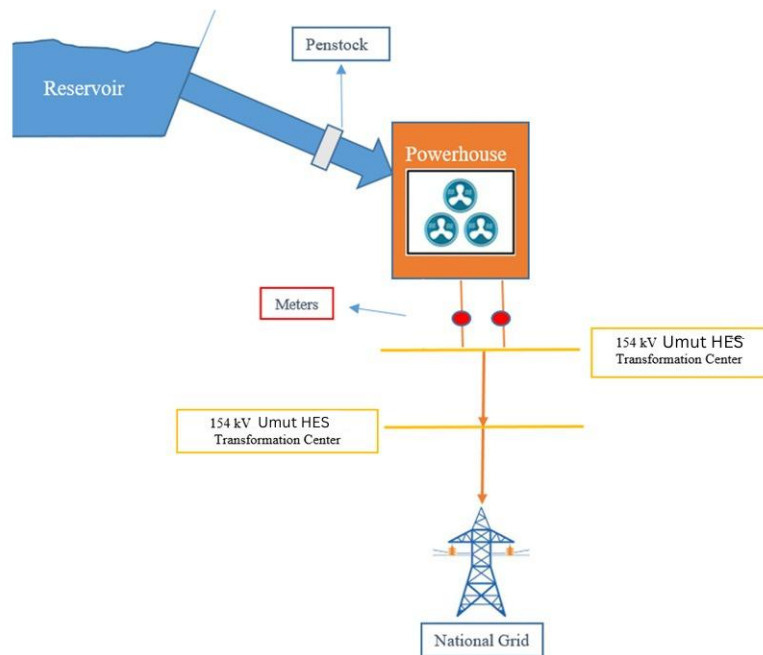


Figure 1. Project Boundary

The spatial extent of the project boundary includes the project power plant, and all power plants connected physically to the national grid. The Project is a renewable energy project, aiming at reducing

GHG emissions resulting from energy generation activities. Therefore, the Project does not involve any GHG sinks or reservoirs. For the identification of the baseline scenario, only electricity generation is considered, as the Project has the sole purpose of generating energy. The sources of baseline GHG emissions are defined as the thermal power plants in Türkiye, which consume a variety of fuels such as naphtha, coal, lignite or natural gas. Plant Load Factor is calculated as below;

$$33.74 \% = 100 \times \frac{72,276 \text{ MWh / year}}{24.45 \text{ MWe}} \times \frac{1}{365 \times 24 \text{ hours}}$$



Figure 1 The reservoir area and the regulator of the Umut II HEPP⁷

1.13 Project Location

The Project is in the Black Sea region of Türkiye, near Akkus town, Ordu Province, Karakuş River. The Powerhouse is located close to the Gokcebayir village, and the diversion weirs are located close to the Karakus village. The locations of the diversion weir and the powerhouse are provided below:

Table 4. Project Coordinates

Coordinate System	Diversion Weirs	Powerhouse
Degrees, Minutes, Seconds (Latitude)	40° 42' 05"	40° 42' 46"
Degrees, Minutes, Seconds (Longitude)	37° 4' 12"	37° 1' 44"

⁷ KML file of Umut HEPP

Decimal (Latitude)	40.701389	40.712778
Decimal (Longitude)	37.07	37.028889



Figure 1. Project Location

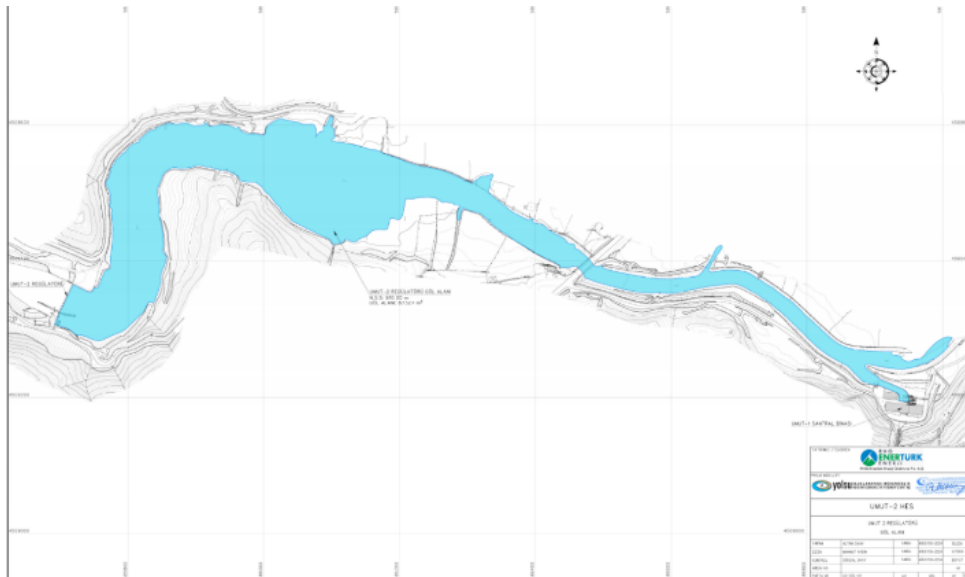


Figure 2. Reservoir Area of the Umut-II HEPP⁸

1.14 Conditions Prior to Project Initiation

⁸ Umut Hes Göl Alanı Document provided by the PP

The Project proponent has no control over the baseline GHG emission sources, which will be substituted by the Project. Therefore, it has not been realized for the purpose of subsequent GHG emission reductions.

Baseline scenario is the same as the conditions existing prior to the project initiation, thus, Section 3.4 will elaborate the baseline scenario.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Türkiye is officially a signatory of the Kyoto Protocol as of 26.08.2009 but has no national legal binding emission reduction targets.

As stated in Section 1.1, EIA Approval is received on 17/12/2009 due to change in capacity of the project. Accordingly, EIA regulation applies on the project. And as stated in Table 6, environment related legal framework apply on the project. The Project follows all required relevant regulations but is not a mandatory Project. There is no breach of any legislation, and all required permits are obtained.

The following laws and regulations apply for power generation facilities in the host country.

Table 5. Legal Framework Concerning Electricity Generation

Legal Framework Concerning Electricity Generation
Electricity Market Balancing and Settlement Regulation
Regulation Concerning Electricity Demand Forecast
Communiqué Regarding Connection to and Use of Transmission and Distribution Systems in the Electricity Market
Communiqué Regarding Preparation of Retail Contract in The Electricity Market
Communiqué Concerning Principles and Procedures for Selection in the Existence of More Than One Application for Generation of Electricity from the Same Source and on the Same Region
Communiqué Regarding Regulation of Market Management Revenue
Regulation on Principles and Procedures for Granting Guarantee of Origin
Electricity Transmission System Supply Reliability and Quality Regulation
Electricity Market Grid Regulation
Communiqué Regarding The Principles and Procedures of Financial Settlement In The Electricity Market

Electricity Market Import and Export Regulation
Communiqué Regarding Regulatory Accounting Guidelines
Electricity Market Distribution Regulation
Communiqué Regarding the Meters to be used in the Electricity Market
Communiqué Regarding Wind and Solar Measurements
Electricity Market Customer Services Regulation
Electricity Market Eligible Consumer Regulation
Electricity Market License Fees
Electricity Market Tariffs Regulation
Electricity Market Licensing Regulation
Electricity Market Law
Other relevant laws:
Law on Utilization of Renewable Energy Resources for the Purpose of Generating Electricity Energy (Nr. 5346)
Energy Efficiency Law (Nr. 5627)
Forest Law (Nr. 6831)

Table 6. Framework Concerning the Environment

Legal Framework Concerning the Environment	
The Environmental Law	Packaging Waste Control Regulation
Air Quality Control Regulation	Soil Protection Law
Noise Control Regulation	Air Pollution from Heating Regulation
Water Pollution Control Regulation	Hazardous Chemicals Regulation
Control of Solid Waste Regulation	Waste Batteries Regulation
Environmental Impact Assessment Regulation	Soil Pollution Control Regulation
Medical Waste Regulation	Air Pollution from Vehicles Regulation

Toxic Chemical Substance and Prod Regulation	Medical Waste Control Regulation
Hazardous Waste Regulation	Water Resources Protection from Hazardous Materials Regulation
Waste Oils Control Regulation	Urban Wastewater Treatment Regulation
Industrial Air Pollution Control Regulation	Drinking Water Quality Regulation

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes

☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18 Sustainable Development Contributions

The project helps with the following sustainable development goals:

SDG 7 (Affordable and Clean Energy): The project promotes clean and sustainable energy production and use. The total net electricity production of the project is estimated to be 722,759 MWh for the whole 2nd crediting period. The annual electricity production of the project is estimated to be 72,276 MWh/year.

SDG 8 (Decent Work and Economic Growth): The project generates employment for all (27 employments for this crediting period). The employees have required technical training and certifications.

SDG 13 (Climate Action): The project contributes to SDG 13 with an estimated amount of 33,013 tons of CO₂e per annum for this crediting period and as total 330,130 tons of CO₂e for the whole 2nd crediting period, which represent direct and quantifiable impact on climate security.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

In accordance with the ACM0002 Version 22.0, leakage is considered as “0” and therefore leakage management is neglected.

1.19.2 Commercially Sensitive Information

The public version of the project description does not exclude any items. During validation of the project description, some confidential documents are provided to the VVB that have not been disclosed in detail in this project description, such as;

- The agreement between the carbon consultant and the project proponent,
- Expert reports prepared by 3rd parties.

1.19.3 Further Information

There is no further information to indicate.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	Through consultation meetings, stakeholders are identified. To identify stakeholders, rules have been followed based on the Environmental Impact Assessment Regulation dating 17.07.2008 and numbered 26939 ⁹ In the EIA Report page 266, stakeholders are defined as the locals who live in the villages near the power plant. ¹⁰
Legal or customary tenure/access rights	In scope of this project, there is no involuntary relocation of households. Local people are mainly interested in fair compensation for their lands. There is minimal private land in the Project region. According to the Turkish Expropriation Laws and Electricity Market Law (4628), court decisions are released indicating that expropriation is finalized without any problem ¹¹ .
Stakeholder diversity and changes over time	No stakeholder diversity and changes over time. Stakeholders are mainly villagers. Therefore, their lifestyles are quite similar over the years. The stakeholder diversity is not changing socially, culturally, and economically over time.
Expected changes in well-being	The project would not have any negative impacts on stakeholders' well-being. For the Umut-II Hydroelectric Power Plant project in Türkiye, the expected changes in well-being relative to the baseline scenario include enhanced local energy security, reduced reliance

⁹ EIA Report, page 266

¹⁰ EIA Report, page 266

¹¹ Please see the EPDK Declarations

	on fossil fuels, and improved air quality due to lower greenhouse gas emissions.
Location of stakeholders	The Project location is a remote location with no direct impact on surrounding villages. Akkus Town, Ordu, Türkiye.
Location of resources	The Project location is a remote location with no direct impact on surrounding villages. Akkus Town, Ordu, Türkiye.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	10-April-2009 at 10.00 AM in Calca Village Niksar, Tokat 10-April-2009 at 14.00 PM in Gokcebayir Village Akkus, Ordu¹² 
---	---

¹² Umut HES Nihai CED Raporu, page 266

	
Stakeholder engagement process	Although the Project location is a remote location with no direct impact on surrounding villages, the closest communities are identified as local stakeholders and are invited. In this meeting, stakeholders have not provided any relevant positive or negative comment on the Project Activity. People are informed about the project and its impacts, which are primarily land acquisition and compensation. The project Proponent has committed to ongoing community engagement activities. “An announcement detailing the location and timing of the meeting was disseminated and publicized in a local newspaper distributed in the Ordu and Tokat provinces (including both the central areas and districts) as well as in a national newspaper to inform the public about the Public Participation Meeting.
Consultation outcome	The project proponent has committed to ongoing community engagement activities, including a secondary stakeholder consultation in 2010. During this consultation, no critical issues were raised by stakeholders concerning the project activity, indicating broad consent for the project design and implementation. However, it is important to note that stakeholder discussions encompassed key aspects such as potential risks, costs, and benefits of the project, as well as compliance with all relevant national laws and regulations, particularly those concerning workers' rights in Türkiye. The discussions also highlighted the application of the Free, Prior, and Informed Consent (FPIC) principle to ensure that local communities were fully informed and had the opportunity to express concerns. Additionally, the project has adhered to the Verified Carbon Standard (VCS) validation and

	verification process, ensuring transparency and accountability in meeting both environmental and social safeguards.
Ongoing communication	Local stakeholders are able to communicate with the project site personnel verbally to share their grievances and inputs in a Continuous Input / Grievance Expression Process Book. A book is provided by PP is available with Chief of HPP to note down Input / Grievance Expression if any local stakeholder would like to share. The grievance might be negative as well as positive can be provided by local stakeholders related to operation of the project activity. PP will resolve all the grievances during whole crediting period.
Stakeholder input	There is no input received from the stakeholders during the consultation.

2.1.3 Free Prior and Informed Consent

Obtaining consent	As a part of the EIA process, to obtain consent of the stakeholders, the community meetings were organized. In these meetings general information about the construction site, duration of the construction and cost of the project, technical information about the equipment's to be used on the project, the potential environmental impacts and the measures to be taken are presented to the stakeholders. After providing all that information to the stakeholders, consent is obtained from stakeholders for the project activity. There are no IPs, LCs or customary rights holders and hence no transparent agreement is reached. Also, there are no ongoing or unresolved conflicts.
Outcome of FPIC	The project proponent helped the public on social issues raised by Stakeholders and provided job opportunities. There are no IPs, LCs or customary rights holders and hence no transparent agreement is reached. Also, there are no ongoing or unresolved conflicts between the PP and the stakeholders. The project has not encroached on land because the land is majorly forest, hence the project has not relocated any people without consent and forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

No grievance was raised during previous crediting period.

Development process	Project manager is accessible by locals all the time. There is no problem encountered about the project/way of
---------------------	--

	communication so far. If there is an input or grievance, locals can express them directly or via mukhtar. PP will resolve all the grievances during the whole crediting period.
Grievance redress procedure	The grievance, negative or positive, can be provided by local stakeholders related to the operation of the project activity. PP will resolve all the grievances during the whole crediting period.

2.1.5 Public Comments

Comments received	Actions taken
No positive or negative comments received from the public.	No actions taken since there is no relevant positive or negative comments received from the public.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

Founded in 2010, RHG Enertürk Enerji Üretim ve Ticaret A.Ş. has a total renewable energy generation portfolio of 208.30 MWe with 8 licensed generation facilities. The management teams have experience in implementing similar project activities and engaging communities. There are 7 HEPPs and 1 WPP managed by the company. Among the HEPPs there are Erenköy HEPP, Yayla HEPP, Sümer HEPP, Tuna HEPP and Yaprak HEPP.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified.	The wellbeing of the stakeholders is required by Environmental Impact Assessment Regulation dating 17.07.2008 and numbered 26939 ¹³ . In this regard, project proponent respects stakeholders and care about their opinions about the Umut-II HEPP Project.

¹³ EIA Report, page 266

Risks to stakeholder participation	There is no risk identified for the stakeholder participation.	The participation of the stakeholders is required by Environmental Impact Assessment Regulation dated 17.07.2008 and numbered 26939 ¹⁴ . In this regard, there cannot be any risks to stakeholders' participation.
Working conditions	There is no risk identified for the working conditions.	Occupational Health and Safety Regulation is implemented in the project to secure the working conditions ¹⁵ .
Safety of women and girls	There is no risk identified for the safety of women and girls.	Türkiye has ratified ILO convention 100 ¹⁶ , 111 ¹⁷ , 122 ¹⁸ and 142 ¹⁹ , which provides gender equality.
Safety of minority and marginalized groups, including children	There is no risk identified for the safety of minority and marginalized groups, including children.	The project tries to align with the national gender strategy. So, the project does not involve and is not complicit in any form of discrimination based on gender difference. All the workers are socially secured by the PP and protected by employment agreements. If any complaint is received by the PO, they will act on the issue right away throughout the lifetime of project activity. PP cannot be complicit in violence or human rights abuses or child/forced labor. Türkiye is a

¹⁴ EIA Report, page 266

¹⁵ EIA Report, page 104

¹⁶ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312245:NO

¹⁷ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312256:NO

¹⁸ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312267:NO

¹⁹ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312287:NO

		party of IPEC ²⁰ since 1932 and ratified ILO convention 29 ²¹ , 138 ²² and 182 ²³ .
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	There is no risk identified for any environment (air, water, soil, etc...)	This project is a renewable energy project and does not pollute any type of environment (air, water, soil). The project received an EIA positive certificate on 17/12/2009 from the Ministry of Environment and Forestry General Directorate of Environmental Impact Assessment and Planning, Türkiye. Any waste formed i.e., discharges, hazardous materials or chemicals are discharged in accordance with the related regulations as per the Environmental Impact Assessment Regulation dated 17.07.2008 and numbered 26939²⁴. As per EIA Report²⁵, Insulating oils are used in the transformer. In case of large-scale malfunctions in the facility, during the oil transfer process to be carried out, as a result of accidents that may occur, absorbent such as sand, gravel or sawdust will be poured on the insulation oil that spills or overflows into the environment to prevent leakage into the

²⁰ <https://www.ilo.org/turkiye>

²¹ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312174:NO

²² https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312283:NO

²³ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312327:NO

²⁴ EIA Report, page 266

²⁵ EIA Report, page 57-58

		environment, and then this mixture will be stored in barrels. Disposal of end-of-life insulation oils will be carried out in licensed disposal facilities as specified in the “Regulation on Control of Waste Oils” published in the Official Gazette dated 30.07.2008 and numbered 26952. Transportation of waste to disposal facilities will be carried out by licensed transporter.
--	--	---

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ²⁶	Mitigation or preventative measure(s) taken
Discrimination	No risk identified.	PP cannot be complicit in violence or human rights abuses or child/forced labor. Türkiye is a party of IPEC ²⁷ since 1932 and ratified ILO convention 29 ²⁸ , 111, 138 ²⁹ and 182 ³⁰ .
Sexual harassment	No risk identified.	
Equal pay for equal work	No risk identified.	Türkiye has ratified ILO convention 100 ³¹ , 111 ³² , 122 ³³
Gender equity in labor and work	No risk identified.	

²⁶ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

²⁷ <https://www.ilo.org/turkiye>

²⁸ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312174:NO

²⁹ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312283:NO

³⁰ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312327:NO

³¹ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312245:NO

³² https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312256:NO

³³ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312267:NO

		and 142³⁴, which provides gender equality. The project tries to align with the national gender strategy. So, the project does not involve and is not complicit in any form of discrimination based on gender difference. The project activity adheres to the regulations related to labor in Türkiye. Labor law: https://turkishlaborlaw.com/category/turkish-labor-law-4857/
Forced labor	No risk identified.	All the workers are socially secured by the PP and protected by employment agreements. If any complaint is received by the PO, they will act on the issue right away throughout the lifetime of project activity. PP cannot be complicit in violence or human rights abuses or child/forced labor. Türkiye is a party of IPEC³⁵ since 1932 and ratified ILO convention 29³⁶, 138³⁷ and 182³⁸. The project activity adheres to the regulations related to labor in Türkiye. Labor law: https://turkishlaborlaw.com/category/turkish-labor-law-4857/
Child labor	No risk identified.	
Human trafficking	No risk identified.	

³⁴ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312287:NO

³⁵ <https://www.ilo.org/turkiye>

³⁶ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312174:NO

³⁷ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312283:NO

³⁸ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312327:NO

2.3.2 Human Rights

PP is never complicit in violence or human rights abuse. If any complaint is received by the PP, they act on the issue right away throughout their lifetime of project activity. Türkiye is a party of IPEC³⁹ since 1932 and ratified ILO convention 29⁴⁰, 138⁴¹ and 182⁴².

Risks identified	Mitigation or preventative measure(s) taken
No risk identified.	PP cannot be complicit in violence or human rights abuses or child/forced labor. Türkiye is a party of IPEC ⁴³ since 1932 and ratified ILO convention 29 ⁴⁴ , 138 ⁴⁵ and 182 ⁴⁶ .

2.3.3 Indigenous Peoples and Cultural Heritage

There is no land which was occupied by IPs. However, the PO recognizes, respects, and promotes the protection of the rights of LCs, and customary rights holders in line with applicable international human rights law. There are no lands which occupy cultural heritage.

Risks identified	Mitigation(s) or preventative measure taken
No risk identified.	There are no lands which was occupied by IPs. PP cannot be complicit in violence or human rights abuses or child/forced labor. Türkiye is a party of IPEC ⁴⁷ since 1932 and ratified ILO convention 29 ⁴⁸ , 138 ⁴⁹ and 182 ⁵⁰ .

³⁹ <https://www.ilo.org/turkiye>

⁴⁰ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312174:NO

⁴¹ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312283:NO

⁴² https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312327:NO

⁴³ <https://www.ilo.org/turkiye>

⁴⁴ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312174:NO

⁴⁵ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312283:NO

⁴⁶ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312327:NO

⁴⁷ <https://www.ilo.org/turkiye>

⁴⁸ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312174:NO

⁴⁹ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312283:NO

⁵⁰ https://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:12100:0::NO:12100:P12100_INSTRUMENT_ID:312327:NO

2.3.4 Property Rights

There are no lands occupied by IP or LCs. In scope of this project, there is no involuntary relocation of households. Local people are mainly interested in fair compensation for their lands. There is minimal private land in the Project region. According to the Turkish Expropriation Laws and Electricity Market Law (4628), court decisions are released indicating that expropriation is finalized without any problem⁵¹.

Risks identified	Mitigation or preventative measure(s) taken
No risk identified.	Local people are mainly interested in fair compensation for their lands. There is minimal private land in the Project region. According to the Turkish Expropriation Laws and Electricity Market Law (4628), court decisions are released indicating that expropriation is finalized without any problem ⁵² .

2.3.5 Benefit Sharing

The project does not affect property rights. So, this section is N/A

Process used to design the benefit sharing plan	N/A
Summary of the benefit sharing plan	N/A
Approval and dissemination of benefit sharing plan	N/A

2.4 Ecosystem Health

Risks identified	Mitigation or preventative measure(s) taken
------------------	---

⁵¹ Please see the EPDK Declarations

⁵² Please see the EPDK Declarations

Impacts on biodiversity and ecosystems	No risk identified.	The project has no impacts on the biodiversity and ecosystems according to the EIA positive decision and the EIA report even though the project involves a reservoir area it does not affect any biodiversity or ecosystem negatively.
Soil degradation and soil erosion	No risk identified.	Erosion control is set by intensifying local characteristics. Soils resistance to erosion was improved. Soils permeability, water pressure formation and seepage is monitored. Site observations are done during visits.
Water consumption and stress	No risk identified.	The plant uses the water left from the channel to generate electricity, before conveying water to the riverbed. The project partially diverts the Karakuş River's water flow to a dam to regulate the water flow and convey it to the powerhouse. The project does not consume the water and releases the water to the channel.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

2.4.2 Introduction of Species

The project does not involve introduction of species; thus, this section is N/A.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
No introduction of species.	-	-	-

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
---------------------------	--

There are no exiting invasive species.	-
--	---

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	No invasive species identified by EIA Report.	-

2.4.3 Ecosystem Conversion

This is not an AFOLU, ARR, ALM, WRC or ACoGS project, so this section is N/A.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	No risk identified.	This is not an AFOLU, ARR, ALM, WRC or ACoGS project. Thus, this is N/A.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	ACM0002 "Consolidated methodology for grid-connected electricity generation from renewable sources"	22.0 ⁵³
Tool	Tool 07	Tool to calculate the emission factor for an electricity system	7.0 ⁵⁴

⁵³ <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

⁵⁴ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

Tool	Tool 01	Tool for the demonstration and assessment of additionality	7.0.0 ⁵⁵
Tool	Tool 11	Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period	3.0.1 ⁵⁶

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s); or (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	The methodology is applicable for renewable energy generation units (incl. hydropower) that supply electricity to a national grid. The methodology is applicable to project activities that install a new, greenfield power plant. Umut II HEPP is a large-scale Greenfield, grid connected renewable electricity generation project. So, the project meets (a) Install a Greenfield power plant.
ACM0002	In case the project activity involves the integration of a	The project activity does not involve integration of a BESS.

⁵⁵ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

⁵⁶ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-11-v3.0.1.pdf>

	BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic¹ or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	Hence, the condition is not applicable.
ACM0002	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or	The project activity includes hydro power with reservoir. Thus, this condition is met.

replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such

	cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	
ACM0002	In case of hydro power plants, one of the following conditions shall apply: 3 (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is	The project activity is a run-of-river type hydro power plant. The project has its own hydromechanical equipment and building. It is implemented in a single reservoir. The project is eligible to apply this methodology since the Power

	implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m² ; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m² ; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m² , and all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m² ; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	density, which is calculated as 280 W/m² is greater than 4 W/m². Thus condition b is applicable to the project. Details of the Power density calculations can be seen in the Section 4.2.
--	---	---

ACM0002	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	Umut II HEPP has its own hydromechanical equipment and building. The HEPP does not involve multiple other HEPPs and the water flowing to the Umut II HEPP is only used by Umut II HEPP. Thus, it is not an integrated HEPP. This condition does not apply to the project.
ACM0002	In the case of PSP, the project participants shall	The project activity does not involve PSP.

	demonstrate in the PDD that the project is not using water which would have been used to generate electricity in the baseline.	
ACM0002	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	(a) The project does not involve switching from fossil fuels to renewable energy sources at the site of the project activity. (b) The project activity does not involve biomass fired power plants/units.
ACM0002	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	The project activity does not involve retrofits, rehabilitations, replacements, or capacity additions.

Applicability as per “Tool 11: Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period, version 3.0.1.

Tool ID	Applicability condition	Justification of compliance
TOOL11	This tool provides a stepwise procedure to assess the continued validity of the baseline and to update the baseline at the renewal of a crediting period, as required by paragraph 49 (a) of the modalities and procedures of the clean development mechanism. The tool consists of two steps. The first step provides an approach to evaluate whether the current baseline is still valid for the next crediting period. The second step provides an approach to update the baseline in case that the current baseline is not valid anymore for the next crediting period	This is the renewal of the crediting period. Therefore, this tool is applicable.

Applicability as per “Tool 07: Tool to calculate the emission factor for an electricity system, version 07.0”.

Tool ID	Applicability condition	Justification of compliance
TOOL07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	The project activity supplies electricity to a grid. Hence, this condition is met.
TOOL07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e., option II a and option IIb. If option II a is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should	CO₂ emission factor for the displacement of electricity generated by power plants in an electricity system is determined by calculating the “combined margin” emission factor (CM) of the electricity system considering only grid-connected plants

Tool ID	Applicability condition	Justification of compliance
	be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
TOOL07	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project electricity system is located totally in an Annex I country. However, this is not a CDM project. Hence, this condition is N/A.
TOOL07	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project does not involve biofuels in any way.

Applicability as per “Tool 01: Tool for the demonstration and assessment of additionality, version 7.0.0,

Tool ID	Applicability condition	Justification of compliance
TOOL01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Project complies with the conditions, this is the renewal of the crediting period, the additionality of the project is demonstrated during the validation step.
TOOL01	Once the additionality tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Project complies with the conditions, this is the renewal of the crediting period, the additionality of the project is demonstrated during the validation step.

3.3 Project Boundary

The spatial extent of the project boundary includes the Project power plant, and all power plants connected physically to the electricity system that the Project power plant is connected to. The Project is a renewable energy project, aiming at reducing GHG emissions resulting from energy generation activities. Therefore, the Project does not involve any GHG sinks or reservoirs.

For the identification of the baseline scenario, only electricity generation is considered, as the Project has the sole purpose of generating energy. The sources of baseline GHG emissions are defined as the thermal power plants in Türkiye, which consume a variety of fuels such as naphtha, coal, lignite or natural gas.

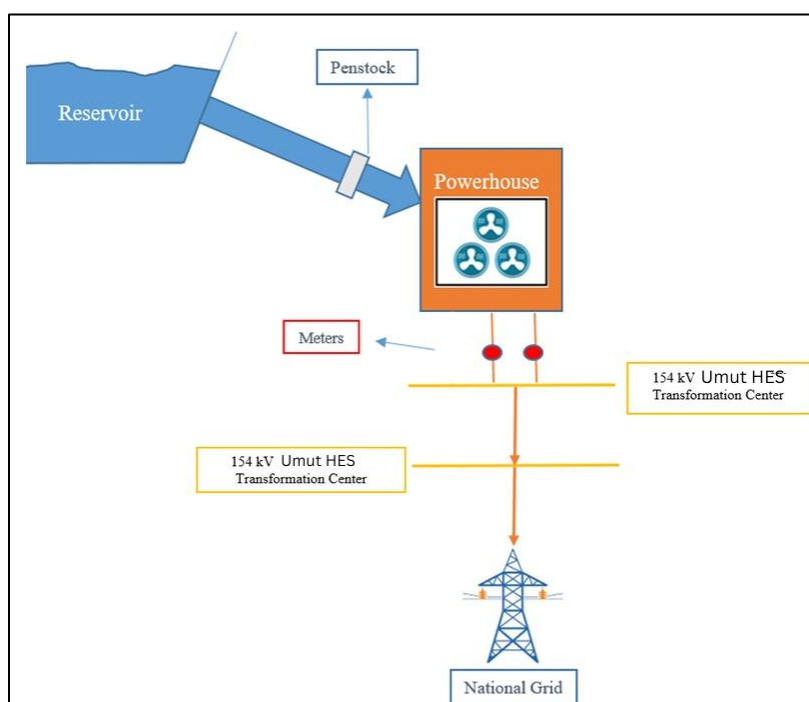


Figure 3. Project Boundary

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Main emission source. The dominant emissions from power plants are in the form of CO ₂ , therefore CO ₂ emissions from fossil fuel fired power plants connected to the grid will be accounted for in baseline calculations.
	CH ₄	No	Minor emission source.

Source		Gas	Included?	Justification/Explanation
		N ₂ O	No	Minor emission source.
Project activity	For dry or flash steam geothermal power plants, emissions of CH ₄ and CO ₂ from non-condensable gases contained in geothermal steam	CO ₂	No	The project activity does not involve dry or flash steam geothermal power.
		CH ₄	No	
		N ₂ O	No	
	For binary geothermal power plants, fugitive emissions of CH ₄ and CO ₂ from noncondensable gases contained in geothermal steam	CO ₂	No	The project activity does not involve geothermal power.
		CH ₄	No	
		N ₂ O	No	
	For binary geothermal power plants, fugitive emissions of hydrocarbons such as nbutane and isopentane (working fluid) are contained in the heat exchangers	Low GWP hydrocarbon/refrigerant	No	The project activity does not involve geothermal power.
	CO ₂ emissions from combustion of fossil fuels for electricity generation in solar thermal power plants and geothermal power plants	CO ₂	No	The project activity does not involve geothermal or solar power.
		CH ₄	No	
		N ₂ O	No	
	For hydro power plants, emissions of CH ₄ from the reservoir	CO ₂	No	The Project does not generate any form of CO ₂ . In urgent cases, a diesel-powered generator will be operated for consumption of the personnel and the building, which is negligible.
		CH ₄	No	The power density of the project activity is greater than 10 W/m ² , hence emissions from water reservoirs are equal to “0” as per paragraph 47 of the applied methodology, ACM0002, V22.0

Source		Gas	Included?	Justification/Explanation
		N ₂ O	No	Minor emission source.
	Charging of BESS using electricity from the grid or from fossil fuel electricity generators.	CO ₂	No	The project activity does not involve BESS.
		CH ₄	No	
		N ₂ O	No	
	Utilization of electricity from grid or from fossil fuel generators by PSP for pumped mode.	CO ₂	No	The project activity does not involve pumped storage.
		CH ₄	No	
		N ₂ O	No	
	For PSP, emissions of CH ₄ from the reservoir	CO ₂	No	The project activity does not involve pumped storage.
		CH ₄	No	
		N ₂ O	No	

3.4 Baseline Scenario

According to applied methodology ACM0002, Version 22.0, paragraph 27, the baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07.

As a condition the project must be a new grid connected renewable energy power plant and the Baseline Emissions are calculated with the “Tool to calculate the Emission Factor for an electricity system” and methodological tool “Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period” and the quantity of net electricity supplied to the grid because of the implementation of the project activity. Identification of the baseline has been made using this approach and is clearly indicated and outlined in the PD.

According to Tool 11: Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period, Version 3.0.1, this is the renewal of the crediting period and there are some requirements:

Step 1: Assess the validity of the current baseline for the next crediting period

Step 1.1: Assess compliance of the current baseline with relevant mandatory national and/or sectoral policies

The current baseline complies with the relevant mandatory national and/or sectoral policies which have come into effect after the submission of the project activity for validation or the submission of the previous request for renewal of the crediting period and are applicable at the time of requesting renewal of the crediting period.

Step 1.2: Assess the impact of circumstances

New circumstances in Türkiye impact the grid's emission factor and, thus, on the project activity's current baseline emissions. Accordingly, the emission factors for the second crediting period have been updated in line with the latest version of the "Türkiye's National Electric Grid Emission Factor" for 2022⁵⁷. There has been no major deviation or change in the market characteristics during the first crediting period.

Step 1.3: Assess whether the continuation of use of current baseline equipment(s) or an investment is the most likely scenario for the crediting period for which renewal is requested.

The expected lifetime of the project is 48 years, 8 months, and 4 days, and the lifetime exceeds the crediting period for which renewal is requested. Equipment in the plant only requires regular maintenance. The baseline scenario, identified at the validation of the project activity, was the continuation of grid-connected electricity generation. Under this scenario, no investment from the project's proponent or a third party(ies) has been envisioned later for the project.

Step 1.4: Assessment of the validity of the data and parameters

Emission reduction calculations are based on two main parameters; energy generation and grid emission factors. The grid emission factor will be updated.

As a methodology requirement, the grid emission factor has been updated during the second crediting period renewal.

Step 2: Update the current baseline and the data and parameters**Step 2.1: Update the current baseline**

According to the methodology, baseline scenario was identified as "the electricity delivered to the grid by the project activity that otherwise would have been generated by the operation of grid connected power plants and by the addition of new generation sources".

Step 2.2.: Update the data and parameters

The emission factor is calculated in accordance with the Tool 07 "Tool to calculate the emission factor for an electricity system", Version 7.0⁵⁸ as follows. In the relevant tool 6 steps are identified to calculate the combined margin (CM) emission factor. Türkiye's National Electricity Network Emission Factor Factsheet

⁵⁷ See Energy and Natural Resources Ministry, Turkish National Electricity Grid Emission Factor Factsheet.
https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Sebeke_EF_Bilgi_Formu_2022.pdf

⁵⁸ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

(2022)⁵⁹ issued by Ministry of Energy and Natural Resources is followed to calculate the CM emission factor. Since the referred document is followed, the steps identified in Tool 07 from Step 1 to Step 5 were already applied by Ministry of Energy and Natural Resources and the followings are calculated:

Parameter	Value	Reference
Operation Margin (OM) emission factor ($EF_{Grid,OM,y}$ as tCO_2e/MWh)	0.7108	Türkiye's National Electricity Network Emission Factor Factsheet (2022)
Build Margin (BM) emission factor ($EF_{Grid,BM,y}$ as tCO_2e/MWh)	0.3721	Türkiye's National Electricity Network Emission Factor Factsheet (2022)
Combined Margin (CM) emission factor ($EF_{Grid,CM,y}$ as tCO_2e/MWh)	The weigh factors for the OM and BM are taken as 0.25 and 0.75, respectively. $CM = 0.25 \times 0.7108 + 0.75 \times 0.3721 = 0.4568$	Tool 07 "Tool to calculate the emission factor for an electricity system", Version 7.0

3.5 Additionality

This is the second crediting period. In the first crediting period, the additionality of the project was presented. There has been no capacity change in the project between the first and the second crediting period. The Project is in compliance with all required relevant regulations but is not a mandatory Project. There is no change in the relevant national and/or sectoral regulations with respect to project activity or its technology till the date through which the project activity is not found mandated by regulations. There is no breach of any legislation, and all required permits are obtained. Laws and regulations that apply for power generation facilities in the host country are mentioned in Section 1.15 of this PD. Hence this project is a regulatory surplus.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☒ Annex 1 country

☐ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

⁵⁹

https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Sebeke_EF_Bilgi_Formu_2022.pdf

☐ Yes

☐ No

The project is not located inside a Non-Annex 1 country and it is not mandated by any law.

3.5.2 Additionality Methods

This is the second crediting period. In the first crediting period, the additionality of the project was presented. There has been no capacity change in the project between the first and the second crediting period. The Project is in compliance with all required relevant regulations but is not a mandatory Project. There is no breach of any legislation, and all required permits are obtained. There is no change in the relevant national and/or sectoral regulations with respect to project activity or its technology till the date through which the project activity is not found mandated by regulations.

3.6 Methodology Deviations

There are no methodological deviations in the project.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Accordingly, the baseline emissions (BE_y) are calculated as following:

$$BE_y = (EG_{PJ,y}) \times EF_{grid,CM,y}$$

where,

BE_y	Baseline emissions (tCO ₂ e)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)
$EF_{grid,CM,y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of TOOL07 (version 7.0) "Tool to calculate the emission factor for an electricity system" (tCO ₂ e/MWh)

Emission factor calculated according to selected methodology and according to the Ministry of Energy and Natural Resources document named as Türkiye's National Electricity Network Emission Factor Factsheet (2022), OM is calculated as 0.7108 tCO₂/MWh whereas BM is 0.3721 tCO₂/MWh. Therefore, CM is calculated as 0.4568 whereas 0.25 and 0.75 weightage factors given to OM and BM, respectively.

Since National Emission Factor calculated by Republic of Türkiye Ministry of Energy and Natural Resources was used, Step-1, Step-2, Step-3, Step-4, and Step-5 was conducted by the Ministry.

Step-6 is applied as follows:

Weighted average CM is calculated using weights, as per Equation 16 Tool 07 version 07.0, paragraph 85:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}$$

Hydroelectric project activities: $w_{OM} = 0.25$ and $w_{BM} = 0.75$ for the crediting period and for subsequent crediting periods.⁶⁰ Hence:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times 0.25 + EF_{grid,BM,y} \times 0.75$$

$$EF_{grid,CM,y} = 0.7108 \times 0.25 + 0.3721 \times 0.75 = \mathbf{0.4568 \text{ tCO}_2/\text{MWh}}$$

Accordingly, BE_y is calculated as follows:

$$BE_y = 72,276 \text{ MWh/y} \times 0.4568 \text{ tCO}_2/\text{MWh} = 33,013 \text{ tCO}_2/\text{y}$$

4.2 Project Emissions

As per ACM0002 version 22.0⁶¹ paragraph 40,

For most renewable energy project activities, $PE_y = 0$. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation 1 in ACM0002:

(a) Emissions related to the operation of geothermal power plants (e.g. noncondensable gases, electricity/fossil fuel consumption);

(b) Emissions from water reservoirs of hydro power plants.

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y} + PE_{PSP,y}$$

where:

PE_y = Project emissions in year y ($\text{tCO}_2\text{e/yr}$)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (tCO_2/yr)

$PE_{GP,y}$ = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y ($\text{t CO}_2\text{e/yr}$)

⁶⁰ Para. 86(a) : <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

⁶¹ <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y (tCO_{2e}/yr)

$PE_{BESS,y}$ = Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (t CO_{2e}/yr)

$PE_{PSP,y}$ = Project emissions from utilizing electricity from the grid for pumping operation of PSP in excess to the production of the renewable power plant operating in coordination with the PSP (t CO_{2e}/yr)

The power density (PD) of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

where:

PD = Power density of the project activity (W/m²)

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W)

Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} = Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m²)

A_{BL} = Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m²). For new reservoirs, this value is zero.

UMUT-II		
Parameter	Description & Unit	Value
PD	Power density of the project activity (W/m ²)	280
CapPJ	Installed capacity of the hydro power plant after the implementation of the project activity (W)	24,450,000
CapBL	Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero	0
APJ	Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m ²)	87,327

ABL	Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m ²).	0
-----	--	---

- Since this project uses hydroelectric power, $PE_{GP,y}$ is “0” (tCO₂/yr)
- Since $PE_{FF,y}$ is used for fossil fuel consumptions, this value is equal to “0” in this HEPP Project. Thus, $PE_{FF,y} = 0$ (tCO_{2e}/yr)
- According to the applied methodology ACM0002 v22.0 para. 47 (c) “If the power density of the project activity is greater than 10 W/m², $PE_{HP,y} = 0$.”
- There is no installation of BESS within the project boundary. Therefore, $PE_{BESS,y}$ is equal to “0”.
- $PE_{PSP,y}$ is equal to 0 because the Umut-II HEPP does not utilize electricity from the grid for pumping operation of PSP in excess to the production of the power plant operating in coordination with the PSP.

Based on those explanations, Project emission (PE_y) is equal to “0”.

4.3 Leakage Emissions

According to applied methodology ACM0002 version 22.0⁶² paragraph 71, no leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g., extraction, processing, transport etc.) are neglected.

Therefore,

$$LE_y = 0$$

As a result, total emission reduction ER_y

$$ER_y = BE_y$$

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Emission reductions are calculated as follows. As per paragraph 72 of applied methodology ACM0002 version 22.0.

⁶² <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

$$ER_y = BE_y - PE_y$$

Equation (17)

Where:

 ER_y = Emission reductions in year y (t CO₂e/yr)

 BE_y = Baseline emissions in year y (t CO₂/yr)

 PE_y = Project emissions in year y (t CO₂e/yr)

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
21 February 2024 – 31 December 2024	28,413	0	0	28,413	0	28,413
01 January 2025 – 31 December 2025	33,013	0	0	33,013	0	33,013
01 January 2026 – 31 December 2026	33,013	0	0	33,013	0	33,013
01 January 2027 – 31 December 2027	33,013	0	0	33,013	0	33,013
01 January 2028 – 31 December 2028	33,013	0	0	33,013	0	33,013
01 January 2029 – 31 December 2029	33,013	0	0	33,013	0	33,013
01 January 2030 – 31 December 2030	33,013	0	0	33,013	0	33,013

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
01 January 2031 – 31 December 2031	33,013	0	0	33,013	0	33,013
01 January 2032 – 31 December 2032	33,013	0	0	33,013	0	33,013
01 January 2033 – 31 December 2033	33,013	0	0	33,013	0	33,013
01 January 2034 – 20 February 2034	4,600	0	0	4,600	0	4,600
Total	330,130	0	0	330,130	0	330,130

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ e / MWh
Description	CO ₂ emission factor of the grid electricity in year y
Source of data	Emission factor is according to the Ministry of Energy and Natural Resources document named as Türkiye's National Electricity Network Emission Factor Factsheet (2022)⁶³. The reference link may not be accessible, the English translation of the document is as follows.

⁶³

https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUES EmisyonFktr/Belgeler/Sebeke_EF_Bilgi_Formu_2022.pdf

51

5.2 Data and Parameters Monitored

Data / Parameter	EGPJ,facility,y																							
Data unit	MWh/year																							
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y																							
Source of data	Electricity meters Primary source: EPIAS screenshots Secondary source: Invoices and TEIAS protocols																							
Description of measurement methods and procedures to be applied	The produced and consumed electricity is continuously measured by the main and spare meters of the power plant. This continuous measurement is recorded in a monthly basis. The collected data will be, once approved by TEIAS, used to calculate the net electricity supplied by the Project. This parameter should be either monitored using bi-directional energy meter or calculated as difference between (a) the quantity of electricity supplied by the project plant/unit to the grid; and (b) the quantity of electricity the project plant/unit from the grid. In case it is calculated then the following parameters shall be measured: (a) The quantity of electricity supplied by the project plant/unit to the grid; and (b) The quantity of electricity delivered to the project plant/unit from the grid																							
Frequency of monitoring/recording	Continuous monitoring and recording. Monthly reporting																							
Value applied	72,276 MWh/year																							
Monitoring equipment	<table><tr><td>Properties</td><td>Main Meter</td><td>Spare Meter</td></tr><tr><td>Brand</td><td>EMH</td><td>EMH</td></tr><tr><td>Type</td><td>LZQJ-XC</td><td>LZQJ-XC</td></tr><tr><td>Class</td><td>1S</td><td>1S</td></tr><tr><td>Serial No</td><td>9420180</td><td>9420181</td></tr><tr><td>Test Date⁶⁶</td><td>23/07/2020</td><td>23/07/2020</td></tr><tr><td>Calibration Date</td><td>14/10/2020</td><td>14/10/2020</td></tr></table>			Properties	Main Meter	Spare Meter	Brand	EMH	EMH	Type	LZQJ-XC	LZQJ-XC	Class	1S	1S	Serial No	9420180	9420181	Test Date ⁶⁶	23/07/2020	23/07/2020	Calibration Date	14/10/2020	14/10/2020
Properties	Main Meter	Spare Meter																						
Brand	EMH	EMH																						
Type	LZQJ-XC	LZQJ-XC																						
Class	1S	1S																						
Serial No	9420180	9420181																						
Test Date ⁶⁶	23/07/2020	23/07/2020																						
Calibration Date	14/10/2020	14/10/2020																						

⁶⁶ Please see the Test Reports file

	<table> <tr> <td>Calibration Due</td> <td>14/10/2030</td> <td>14/10/2030</td> </tr> </table>	Calibration Due	14/10/2030	14/10/2030
Calibration Due	14/10/2030	14/10/2030		
QA/QC procedures to be applied	There will be no sampling and all measured data will be used for monitoring. Measurement equipment will be calibrated and their accuracy will be ensured by TEIAS. Both measurement devices will be approved by TEIAS, and the calibration will also be managed by TEIAS. According to the “ÖLÇÜ VE ÖLÇÜ ALETLERİ MUAYENE YÖNETMELİĞİ” / Article 9 the periodic calibration shall be made every 10 years⁶⁷. Both the primary and secondary devices are sealed during first operation of the plant. In case there is a significant inconsistency (assessed by TEIAS) between two devices, the meters will be calibrated by TEIAS before the 10 years period. The calibration due dates of the both meters is 14/10/2030. The Project Proponent can also request calibration of the meter(s). The calibration stations are inspected by the central organization of the same Ministry. Both metering instruments are sealed and only accessible by TEIAS personnel. EPIAS data is also of high quality, as the data is used for invoicing purposes and is accessible online by the Project Proponent, minimizing the risks of data corruption, data archiving, loss of information etc.			
Purpose of data	To calculate the quantity of net electricity delivered to the grid			
Calculation method	$EG_{Facility,y}=EG_{export,y}-EG_{import,y}$ (Annual electricity generation of the plant is calculated as the sum of the monthly generations obtained from EPIAS records. After deducting the amount of imported electricity from the generated electricity, the net electricity supplied to the grid is evaluated.) UMUT-I HYDROPOWER PLANT, Türkiye & UMUT-II HYDROPOWER PLANT, Türkiye & UMUT-III HYDROPOWER PLANT, Türkiye are connected to the same electricity meters. Hence, their electricity generation data are read from one source. Their installed capacities are different, the ratio of their installed capacities are used to divide the electricity generation to each of the plants. It is not possible to monitor electricity generation values separately.Installed capacities from the generation licence are as follows. Installed capacity of UMUT-I HPP: $2 \times 3.037 \text{ MWm} / 2 \times 2.900 \text{ MWe} = 6.074 \text{ MWm} / 5.800 \text{ MWe}$			

⁶⁷ <https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=6381&MevzuatTur=7&MevzuatTertip=5>

Installed capacity of UMUT-II HPP: 3x8.435 MWm / 3x8.150 MWe
= 25.305 MWm / 24.450 MWe

Installed capacity of UMUT-III HPP: 3x4.255 MWm / 3x4.000 MWe
= 12.765 MWm / 12.000 MWe

The electricity generations for each plant is calculated as follows

$$\begin{aligned} & \text{Generation of UMUT I HYDROPOWER PLANT, TURKEY} \\ & = \frac{\text{Total generation value} \times \text{Installed capacity of UMUT I}}{\text{Total installed capacity}} \end{aligned}$$

$$\begin{aligned} & \text{Generation of UMUT II HYDROPOWER PLANT, TURKEY} \\ & = \frac{\text{Total generation value} \times \text{Installed capacity of UMUT II}}{\text{Total installed capacity}} \end{aligned}$$

$$\begin{aligned} & \text{Generation of UMUT III HYDROPOWER PLANT, TURKEY} \\ & = \frac{\text{Total generation value} \times \text{Installed capacity of UMUT III}}{\text{Total installed capacity}} \end{aligned}$$

Consider if the electricity generation in year x has been 124,894 MWh, read from the meter which all the plants are connected.

$$\begin{aligned} & \text{Generation of UMUT I HYDROPOWER PLANT, TURKEY} \\ & = 124,894 \text{ MWh} \times \frac{5.8 \text{ MWe}}{42.250 \text{ MWe}} \\ & = 17,145.21 \text{ MWh} \end{aligned}$$

$$\begin{aligned} & \text{Generation of UMUT II HYDROPOWER PLANT, TURKEY} \\ & = 124,894 \text{ MWh} \times \frac{24.450 \text{ MWe}}{42.250 \text{ MWe}} \\ & = 72,275.94 \text{ MWh} \end{aligned}$$

$$\begin{aligned} & \text{Generation of UMUT III HYDROPOWER PLANT, TURKEY} \\ & = 124,894 \text{ MWh} \times \frac{12 \text{ MWe}}{42.250 \text{ MWe}} \\ & = 35,472.85 \text{ MWh} \end{aligned}$$

$$\begin{aligned} & \text{Total generation} \\ & = 17,145.21 \text{ MWh} + 72,275.93 \text{ MWh} \\ & + 35,472.85 \text{ MWh} = 124,893.99 \text{ MWh} \\ & \approx 124,894 \text{ MWh} \end{aligned}$$

	During verification, “net generation value” will replace “total generation value”. Where, (Net Generation) = (Gross Generation) – (Electricity consumed from grid) Total generation value from the generation licence is only considered here for estimation purpose only.
Comments	-

5.3 Monitoring Plan

The GHG emission reductions are achieved through electricity generation by the Project. Therefore, the net electricity supplied by the Project, which is the difference of gross electricity generation and Project’s own electricity consumption, will be monitored. For this purpose, continuously collected electricity generation and consumption data will be reported and recorded monthly. The measured data are the monthly electricity generation and consumption values. This data is read and reported by TEIAS and can be called from the EPIAS⁶⁸ web site during the following month.

There are two electricity meters (namely main and check meters), which have a brand name EMH and type LZQJ-XC, on site that continuously monitor electricity generation and consumption data and record them digitally. These readings can be called from the SCADA system, but the Project proponent has no means to manipulate the readings or the measurement devices, as they will be sealed by TEIAS. On the first days of each successive month, power generation and consumption readings from both measurement devices will be read by TEIAS and a monthly protocol are prepared based on these readings. This protocol is uploaded to the EPIAS web site, where the Project proponent can easily access and read daily and monthly power reports.

Electricity meters are approved by TEIAS, and the calibration is managed by TEIAS. According to the “ÖLÇÜ VE ÖLÇÜ ALETLERİ MUAYENE YÖNETMELİĞİ” / Article 9 the periodic calibration shall be made every 10 years⁶⁹. Both the primary and secondary devices are sealed during first operation of the plant. In case there is a significant inconsistency (assessed by TEIAS) between two devices, the meters will be calibrated by TEIAS before the 10 years period. The calibration due dates of the both meters is 14/10/2030. The Project Proponent can also request calibration of the meter(s). The calibration stations are inspected by the central organization of the same Ministry.

The electricity generation and consumption data that are recorded and reported during the monitoring period will be compiled with a spreadsheet program for verification purposes. This

⁶⁸ <https://www.epias.com.tr/epias-kurumsal/tarihce/#:~:text=18%20Mart%202015%20tarihinde%20kurulan,i%C5%9Fletilmesi%20ve%20geli%C5%9Ftirilmesi%20i%C3%A7in%20%C3%A7al%C4%B1%C5%9Fmaktad%C4%B1r.&text=Organize%20YEK%2DG%20Piyasas%C4%B1%20ilk%20seans%C4%B1%20ger%C3%A7ekle%C5%9Fti.&text=Vadeli%20Elektrik%20Piyasas%C4%B1%20a%C3%A7%C4%B1ld%C4%B1>

⁶⁹ <https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=6381&MevzuatTur=7&MevzuatTertip=5>

spreadsheet, along with documentary evidence, will be provided to verifying VVB for verification. The sample monitoring spreadsheet is also provided to the validating VVB.

The following procedure will be used to calculate the emission reductions:

The measurement instruments give two types of data (besides others); monthly gross electricity generated, and monthly electricity consumed by the Project. The difference between these two parameters is the monthly net electricity supplied to the grid. These monthly net electricity supply figures will be added up for all months during the monitoring period and the sum will be multiplied with the ex-ante CM emission factor to find the amount of emissions reductions. All the data will be electronically archived and communicated, and the calculations will be reported in a separate spreadsheet for verification purposes. The calculations will be automatically done by the spreadsheet file with minimal human interaction.

The primary source of data is the EPIAS web site, which is the basis for financial settlement and invoicing. The Project Proponent has an access code for its own projects to the EPIAS web site and can call, report and print actual and historical electricity generation and consumption data of the Project activity. These EPIAS reports can only be obtained by the Project Proponent, therefore screenshots will also be provided during verification for QA/QC reasons.

In case there is any unforeseen problem with the EPIAS data, the secondary source of data are the invoices. In this case, the invoices shall be cross checked with the monthly measurement protocols, which are read from the primary and secondary electricity meters by TEIAS and signed by TEIAS and the Project Proponent's responsible person.

If any other unforeseen problem occurs, the electricity generation and consumption data used for invoicing or settlement with TEIAS will be valid.

The measurement devices measure and record data continuously and these readings can be monitored online by the Project Proponent's SCADA system. These data will be collected and reported once a month by TEIAS, and this measurement protocol is uploaded to the EPIAS web site for invoicing purposes. The monitoring of emission reductions will also be carried out on a monthly basis, parallel to electricity generation reporting. Therefore, the frequency of monitoring is continuous, and the frequency of reporting is monthly. The collected data will also be kept by the Project Proponent for at least two years after the credit period.

The environmental engineer of the Project Proponent is responsible for the preparation of the monitoring report. The Project Proponent is responsible for assigning personnel as the carbon manager, whose tasks include:

- Collecting and archiving monthly EPIAS reports and screenshots electronically and as hardcopy,
- Collecting and archiving monthly invoices and TEIAS protocols electronically and as hardcopy,

- Reporting in written form and collecting documentary evidence about any unusual issue regarding protocols, invoices, EPIAS web site access, settlements, technical problems that might affect data quality and availability, significant equipment repair or maintenance and calibration, failure or replacement of measurement devices.
- Ensuring that information flow about the issues mentioned above between the Project manager on site and the carbon manager runs continuously during the whole monitoring period.
- Prepare the monitoring report or provide the collected data to any assigned 3rd party that is responsible for supporting the verification of emission reductions.

Before installation, metering devices are technically checked, calibrated and certified by independent companies for their accuracy. Afterwards, TEIAS also tests both metering devices to ensure that the devices satisfy all requirements and are calibrated properly. In case of failure of the electricity meters or if any difference occurs between primary and secondary devices TEIAS has to be informed by the Project Proponent for necessary maintenance and calibration. An agreement between the project proponent and TEIAS will ensure that in case of problems or failures of the meters TEIAS will react as fast as possible to solve the problem and perform necessary maintenance and calibration. Since the Project Proponent will invoice TEIAS for the electricity it sells to the grid, the data is of high quality. If it is determined by TEIAS that the seal of the primary metering device is torn apart or any intervention is made; correct power amounts are determined over the secondary meter, beginning with the latest measurement when the recorded values of the primary and the secondary metering devices match. If any difference is seen by the Project Proponent between primary and secondary devices, TEIAS will be informed accordingly.

Calibration of measurement devices is the responsibility of TEIAS, and it is managed by the same authority, without Project Proponent's control according to the current legislation. If the responsibilities and legal requirements about calibration change, the Project Proponent will act accordingly. Since monitoring electricity generation and consumption is the responsibility of TEIAS, the Project Proponent has little control over how, where and when TEIAS makes the readings. In case the electricity data cannot or does not get recorded by TEIAS on site, but rather on the nearest transformer station for any unknown reason (e.g. insufficient road access to the Project due to weather conditions etc., data transmission problem etc.), protocols might not be available.

Under normal circumstances, the protocols are signed monthly on site by TEIAS and the plant manager. After the protocol, internal SCADA data are cross checked with the protocol for quality control purposes. Usually, information in both sources should be in agreement. However, should the data from both sources indicate a mismatch for any reason, then the Project Proponent may apply to TEIAS for correction of data and financial settlement. These cases will be reported in the monitoring report and if any settlement takes place the final figures will be used for calculations and documentary evidence will be provided to the verifying DOE.

Other non-conformities regarding readings, calibration and other TEIAS are subject to the bilateral communication and settlements between TEIAS and the Project Proponent. In such a case, the results of these settlements will be used as the valid data source.

VER Team Members is expected to include the following staff of the HEPP:

Plant Engineers (Electrical and mechanical): Responsible for the specific actions required by the monitoring plan; control of the electricity supplied to the grid and imported from the grid with TEIAS, and of devices.

Accounting Manager: Responsible for keeping data about power sales, invoicing and purchasing.

Operations Manager: the VER coordinator, responsible for developing, executing, analysing and improving the VER Monitoring/Reporting Procedures.

GTE Karbon Sürdürülebilir Enerji Eğitim Danışmanlık ve Ticaret A.Ş.: Responsible for emission reduction calculations, preparing monitoring report and periodical verification process.

Each period, the monitoring report is submitted to VVB for the verification. The report covers the monitoring of grid-connected power generation, check report, report on calculation of emission reductions and records of monitoring instrument repair and calibration, etc. All data collected as part of monitoring is archived electronically and kept for at least for 2 years after the end of the last crediting period. The company established a dedicated maintenance system to ensure the data availability for the required period.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

There is no commercially sensitive information given in the project description.

Section	Information	Justification
NA	NA	NA

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.